

CEMAC Jobs Analytics

Cameroon elasticities and WBES trade diagnostics

May 25, 2026

Roadmap

1. BDF Cleaning Results
2. Cameroun Census data on firm counts, employment, and revenue
3. Cameroun Sectoral Elasticities
4. CEMAC WBES Trade Analysis

Data cleaning overview

- The working panel is organized at the firm-year level, so duplicate handling changes both sample size and identification.
- Economic sector is based on the NACAM variable, which according to documentation its based on a Regional harmonized classifier linked to ISIC.
- The cleaning note isolates the two steps that most affect downstream interpretation: duplicate resolution and the NACAM-to-ISIC crosswalk.

Firm-Year Duplicate Cleaning

- Working key: `firmid × fin_yr`
- Perfect duplicates would be collapsed with original row order as the tie-breaker.
- Conflicting duplicate firm-years are fully removed from the cleaned analysis file, while we figure out the best approach for handling them.

	Value
Original observations	9,325
Perfect duplicate firm-years	0
Rows dropped from perfect duplicates	0
Conflicting duplicate firm-years	127
Rows dropped from conflicting duplicates	257
Final cleaned observations	9,068

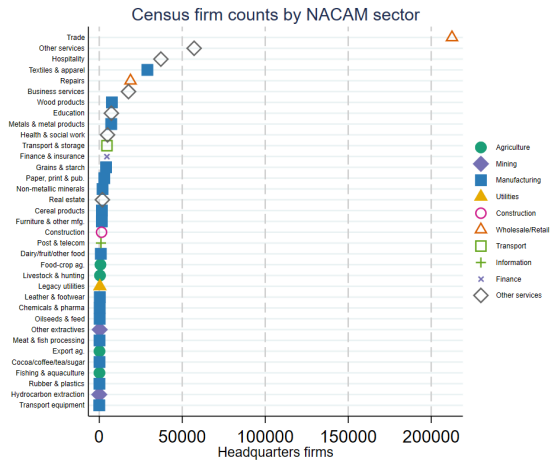
Cleaning Takeaways

- The cleaned panel resizes the sample from 9,325 to 9,068 observations after removing duplicate firm-years.
- There are 127 conflicting firm-years, accounting for 257 dropped obs. Those observations are excluded rather than resolved ad hoc.
- To avoid introducing unwanted noise we keep the analysis using NACAM classification and link it to ISIC aggregates when possible.

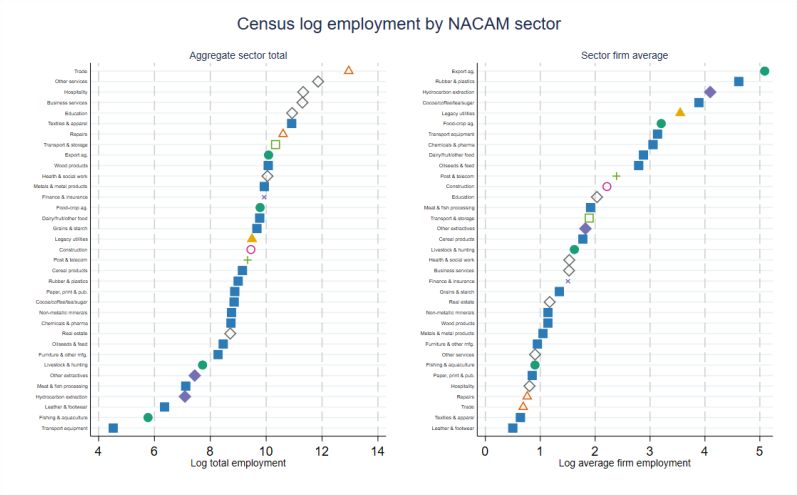
Cameroun Census Context

- The 2024 RGE census gives a whole-economy snapshot of establishments in Cameroon.
- Census activity labels are harmonized to the same NACAM sectors used in the administrative elasticity analysis.
- The next slides use headquarter-level data to show sector scale in firm counts, log employment, log annual revenue, and log revenue per worker.
- After this economy-wide view, the administrative tax panel focuses on formal firms observed over time, using the administrative data.

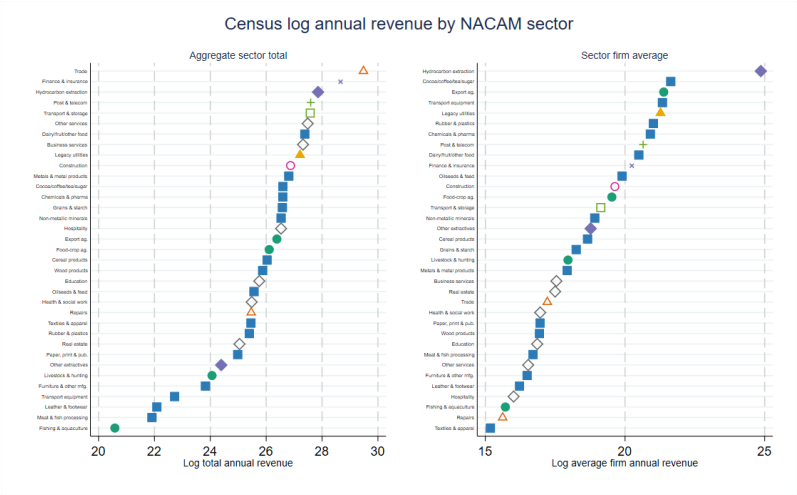
Census Firm Counts by Harmonized NACAM Sector



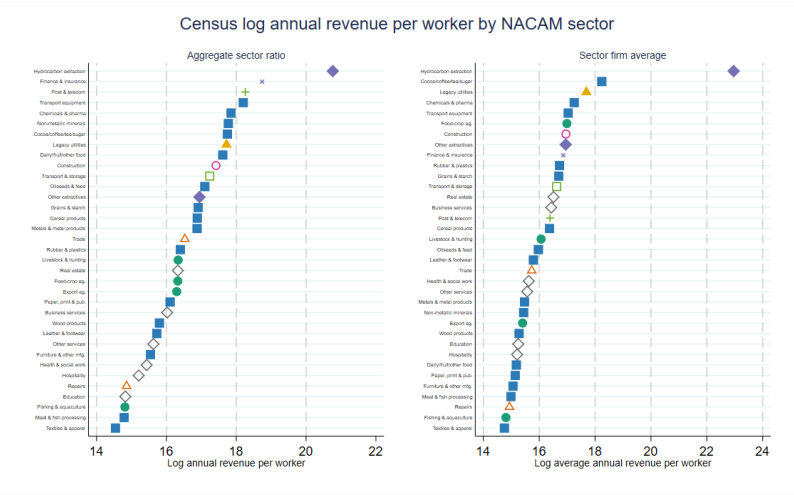
Census Log Employment by Harmonized NACAM Sector



Census Log Revenue by Harmonized NACAM Sector



Census Log Revenue per Worker by Harmonized NACAM Sector

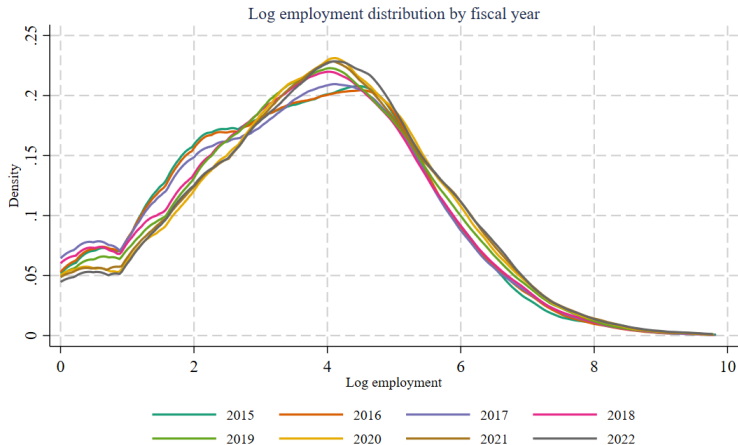


Specification and Sample

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s (\ln(X_{it}) \times D_{is}) + \varepsilon_{it} \quad (1)$$

- The unit of observation is a firm-year (i, t) .
- E_{it} is employment for firm i in year t .
- X_{it} is either value added or total revenue, estimated in separate regressions.
- D_{is} equals 1 if firm i belongs to NACAM sector s , and 0 otherwise.
- Sector is treated as a firm characteristic, so each firm contributes to one NACAM interaction term.
- α_i are firm fixed effects and δ_{st} are sector-by-year effects.
- β_s is the elasticity for firms in sector s , identified through the interaction between $\ln(X_{it})$ and the sector indicator D_{is} .
- Standard errors clustered at the firm level.
- Inclusion rule for reported sectors: at least 30 observations and 10 firms in both outcome samples.

Employment Distribution by Fiscal Year

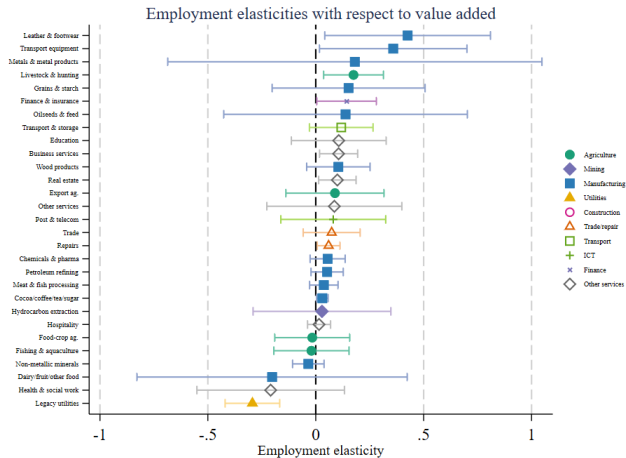


Value-Added Elasticity Specification

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s^{VA} (\ln(VA_{it}) \times D_{is}) + \varepsilon_{it} \quad (2)$$

- Estimates use the Cameroon administrative firm-year panel.
- Firm fixed effects absorb time-invariant firm differences.
- Sector-by-year effects absorb shocks common to a NACAM sector in a fiscal year.
- The plotted point for each sector is β_s^{VA} with a firm-clustered 95 percent interval.

Value-Added Elasticities Across NACAM Sectors

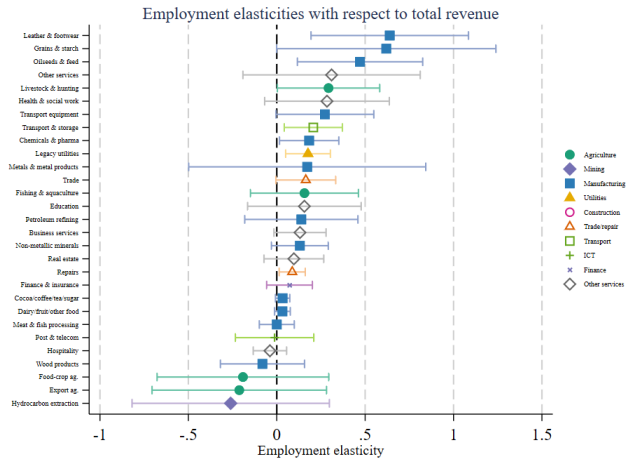


Total-Revenue Elasticity Specification

$$\ln(E_{it}) = \alpha_i + \delta_{st} + \sum_s \beta_s^R (\ln(R_{it}) \times D_{is}) + \varepsilon_{it} \quad (3)$$

- R_{it} is total revenue, estimated in a separate regression from value added.
- The sample rule remains at least 30 observations and 10 firms in both reported outcome samples.
- The plotted point for each sector is β_s^R with standard errors clustered at the firm level.

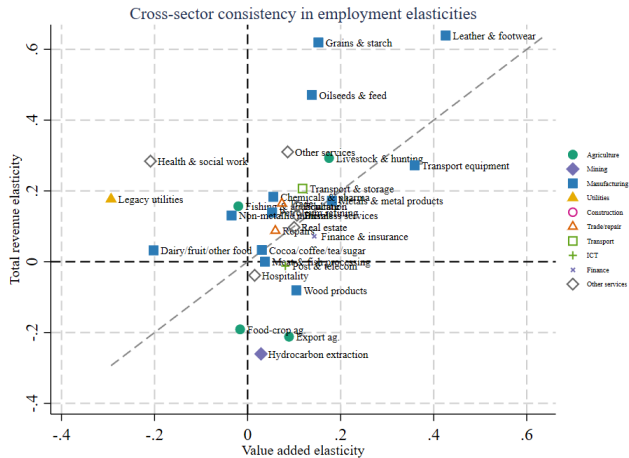
Total-Revenue Elasticities Across NACAM Sectors



Cross-Model Comparison

- The next plot compares the sector-specific value-added and total-revenue elasticities.
- Each point is one NACAM sector retained by the common inclusion rule.
- Points near the 45-degree line indicate sectors where both revenue proxies give similar employment responses.
- Large gaps flag sectors where value added and gross revenue tell different stories.

Cross-Model Consistency

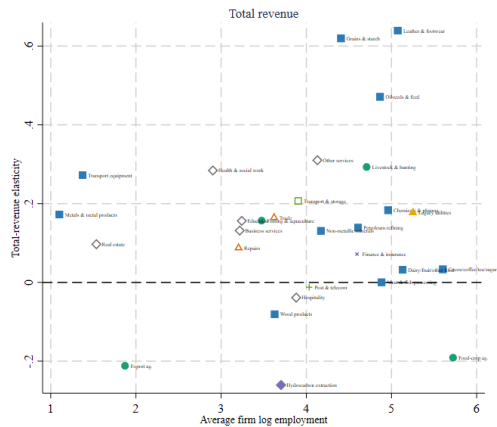
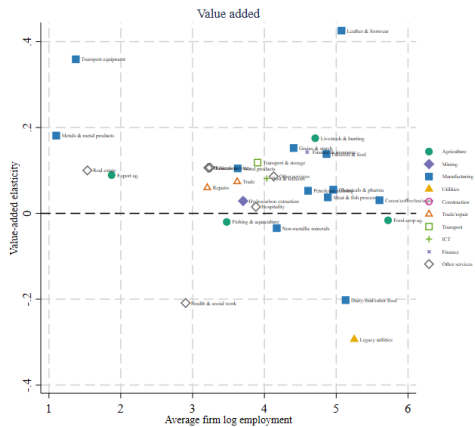


Elasticity and Scale Comparison

- The next scatter plots reuse the same estimated sector elasticities.
- Sector scale is calculated from the common valid sample before plotting.
- Employment scale uses average firm log employment within sector.
- Revenue scale uses average firm log total revenue within sector.

Elasticities and Employment Scale

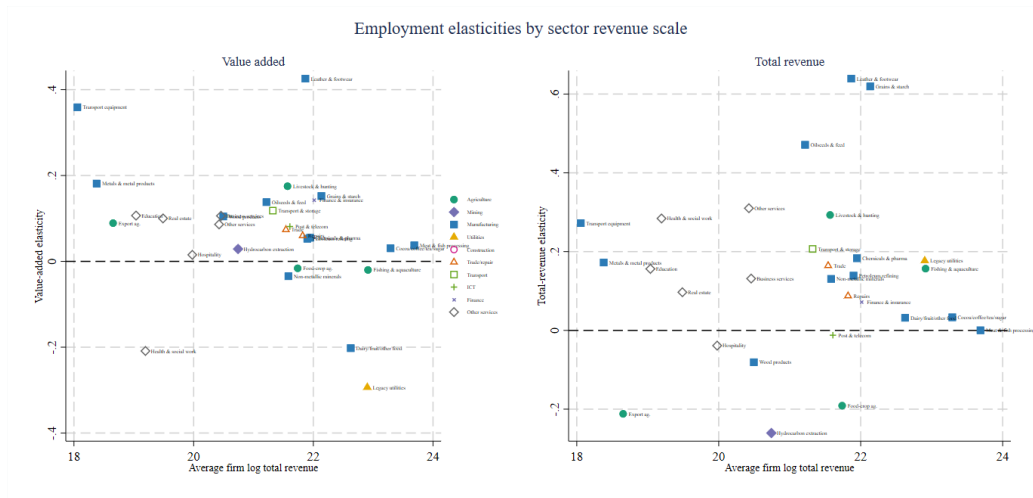
Employment elasticities by sector employment scale



Elasticity and Revenue Scale Comparison

- This plot asks whether high-elasticity sectors are also large-revenue sectors.
- Sector labels and included sectors match the administrative elasticity outputs.

Elasticities and Revenue Scale



Revenue-Elasticity Sector Deciles

	Revenue decile	Total revenue elasticity	Value added elasticity	Revenue observations	VA observations
Leather & footwear	10	0.639	0.426	164	148
Grains & starch	10	0.620	0.152	78	67
Oilseeds & feed	9	0.471	0.138	128	115
Other services	9	0.310	0.086	60	55
Livestock & hunting	9	0.293	0.175	235	197
Health & social work	8	0.284	-0.209	263	243
Transport equipment	8	0.272	0.359	145	125
Transport & storage	8	0.207	0.118	788	708
Chemicals & pharma	7	0.183	0.055	110	100
Legacy utilities	7	0.177	-0.294	56	54
Metals & metal products	7	0.172	0.181	53	45
Trade	6	0.164	0.074	654	575
Fishing & aquaculture	6	0.157	-0.020	79	71
Education	6	0.156	0.107	368	364
Petroleum refining	5	0.139	0.053	215	196
Business services	5	0.131	0.106	1,387	1,264
Non-metallic minerals	5	0.131	-0.035	222	183
Real estate	4	0.097	0.100	325	301
Repairs	4	0.088	0.060	1,712	1,522
Finance & insurance	4	0.072	0.143	300	272
Cocoa/coffee/tea/sugar	3	0.033	0.031	70	57
Dairy/fruit/other food	3	0.032	-0.202	76	64
Meat & fish processing	3	0.000	0.037	65	43
Post & telecom	2	-0.012	0.081	281	258
Hospitality	2	-0.039	0.015	241	224
Wood products	2	-0.081	0.105	183	171
Food-crop ag.	1	-0.191	-0.016	54	51
Export ag.	1	-0.212	0.089	174	134
Hydrocarbon extraction	1	-0.260	0.029	59	48

Decile 10 is highest; decile 1 is lowest.

Main Takeaways

- Elasticities vary meaningfully across sectors rather than lining up around a single Cameroon-wide coefficient.
- Negative or weak value-added elasticities remain concentrated in a small set of sectors, which is useful for prioritizing follow-up diagnostics.

Phase 3: WBES Trade Diagnostics

- Phase 3 uses the latest available World Bank Enterprise Survey wave for CEMAC countries.
- The first step is a cleaning and availability audit for sales, employment, sector, export, import, and input-origin variables.
- Gabon is not retained in the main descriptive panels because its latest available WBES wave is 2009 and has no ISIC Rev.4 coverage.
- The WBES formal-sector sampling frame covers non-agricultural private firms, so sector rows use eligible activity aggregates rather than agriculture.

WBES Descriptive Statistics

	Cameroon	Central Afr. Rep.	Chad	Congo	Eq. Guinea	SSA excl. CEMAC	High-income countries
Sample composition							
Fiscal year	2023	2022	2022	2023	2023	2022	2023
Surveyed firms	615	151	164	365	173	18252	30855
Firms represented (000s)	5	5	1	2	1	354	6879
	14.6	28.9	5.3	12.9	15.8	11.9	18.8
% Exporting firms	[2.1]	[6.9]	[2.5]	[2.8]	[3.6]	[0.6]	[0.6]
	17.7	15.2	28.9	29.8	43.3	24.6	16.3
% Manufacturing	[2.7]	[5.1]	[4.4]	[3.6]	[4.5]	[0.8]	[0.5]
	8.8	15.0	3.1	4.0	7.8	10.0	15.1
% Construction/utilities	[1.7]	[5.5]	[2.3]	[1.3]	[2.7]	[0.7]	[0.6]
	61.7	67.3	59.5	61.4	34.0	60.9	53.3
% Trade/hospitality/transport	[3.3]	[6.8]	[5.3]	[3.9]	[4.6]	[1.0]	[0.8]
	11.9	2.5	8.5	4.8	14.8	4.5	15.4
% Other services	[2.0]	[1.0]	[3.5]	[1.9]	[3.2]	[0.4]	[0.7]
Economic activity (% total sales)							
	17.2	15.6	16.7	11.8	27.6	31.6	24.4
Manufacturing	[3.2]	[5.1]	[5.7]	[3.9]	[6.6]	[2.6]	[1.2]
	10.5	23.0	3.5	5.9	25.8	16.1	14.1
Construction/utilities	[3.3]	[11.6]	[2.4]	[2.8]	[9.4]	[2.4]	[1.4]
	59.7	45.9	70.1	78.6	35.1	48.8	45.7
Trade/hospitality/transport	[5.5]	[10.0]	[7.8]	[6.2]	[8.9]	[2.7]	[1.8]
	12.6	15.4	9.7	3.7	11.5	3.6	15.8
Other services	[4.4]	[10.8]	[4.4]	[3.4]	[6.2]	[0.7]	[1.5]
Average sales (current USD thousands)							
	447.3	36.0	312.5	1058.7	1075.0	763.3	6325.6
All firms	[44.3]	[6.9]	[53.1]	[226.4]	[184.5]	[38.3]	[209.6]
	795.4	47.1	894.1	3588.8	1002.5	1771.2	12779.7
Exporting firms	[166.3]	[19.1]	[444.0]	[1439.2]	[522.5]	[164.6]	[631.7]
	387.4	31.9	280.2	726.7	1088.0	633.6	4801.6
Non-exporting firms	[43.6]	[7.3]	[41.5]	[164.6]	[197.2]	[36.8]	[208.5]
Average employment							
	27.8	22.3	15.2	40.9	32.0	44.9	39.2
All firms	[2.8]	[5.3]	[1.6]	[9.0]	[4.4]	[4.9]	[4.3]
	51.3	14.7	23.0	151.5	19.4	93.5	73.8
Exporting firms	[7.3]	[2.9]	[6.6]	[57.0]	[7.4]	[19.5]	[13.8]
	23.8	24.5	14.7	26.4	34.4	38.9	31.1
Non-exporting firms	[3.1]	[8.3]	[1.7]	[4.7]	[4.9]	[5.1]	[4.2]

Sales use fiscal-year official exchange rates. Brackets report standard errors.

WBES Country Coverage

	Firms	SurveyYear	FiscalYear	FirmsWithISIC4	Retained
Cameroon	615	2,024	2,023	615	1
Central African Republic	151	2,023	2,022	151	1
Chad	164	2,023	2,022	164	1
Congo	365	2,024	2,023	365	1
Equatorial Guinea	173	2,024	2,023	173	1
Gabon	179	2,009	.	0	0

Retained excludes Gabon from the main descriptive panels because its latest available wave is 2009 and lacks ISIC Rev.4.

Core Variable Availability

	Firms	Sales	SalesW	Employment	Weight	ISIC4	ExportShare	ImportStatus	ImportValue	SalesShareOff100	InputShareOff100
Cameroon	615	614	614	615	615	615	615	613	275	0	0
Central African Republic	151	139	139	151	151	151	134	126	56	0	0
Chad	164	164	164	164	164	164	164	164	110	0	0
Congo	365	352	352	358	365	365	353	331	123	0	0
Equatorial Guinea	173	172	172	173	173	173	173	173	108	0	0

Counts are unweighted establishments in retained CEMAC latest-wave surveys.

Negative Special-Code Audit

	Employment	PermanentEmp	Sales	DomesticShare	IndirectExport	DirectExport	DomesticInput	ForeignInput	DirectImport	RawMatCost	ResaleCost
Cameroon	0	0	1	0	0	0	2	2	1	12	1
Central African Republic	0	0	12	11	16	14	25	25	19	3	0
Chad	0	0	0	0	0	0	0	0	0	0	0
Congo	0	7	13	6	10	10	34	34	1	27	42
Equatorial Guinea	0	0	1	0	0	0	0	0	0	0	0

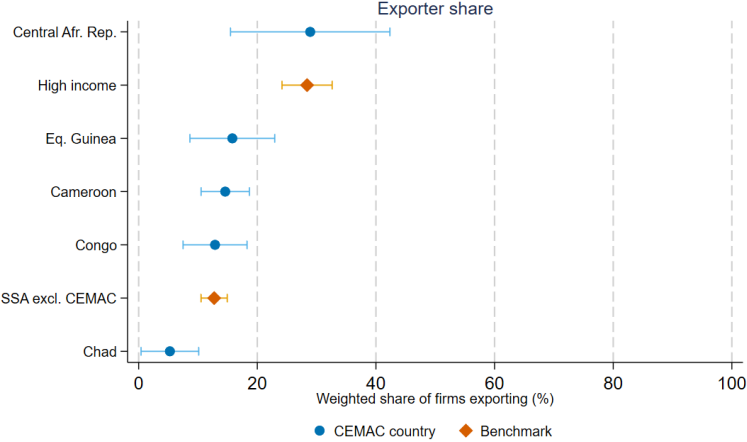
Negative WBES special codes are audited before being recoded to missing for nonnegative levels, percentages, and binary responses.

Winsorization Cutoffs

	SalesP5	SalesP95	InputCostP5	InputCostP95
Cameroon	9,125,300	2930040000	2,000,000	2800000000
Central African Republic	1,744,000	435,000,000	300,000	50,000,000
Chad	6,750,000	1115225000	1,380,000	500,000,000
Congo	1,000,000	12000000000	150,000	600,000,000
Equatorial Guinea	13,000,000	9090000000	300,000	669,000,000

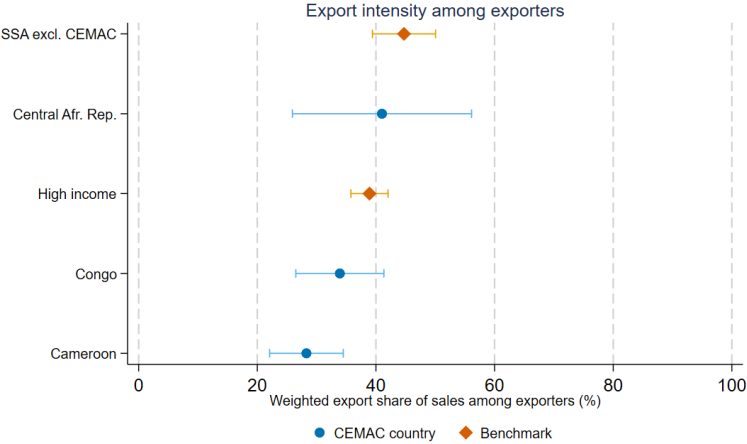
Sales and input costs are winsorized at the 5th and 95th percentiles within each country wave.

Exporter Participation



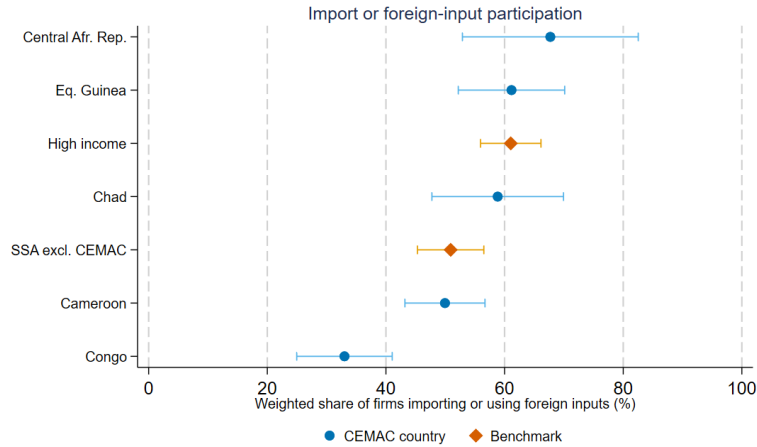
WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

Export Intensity Among Exporters



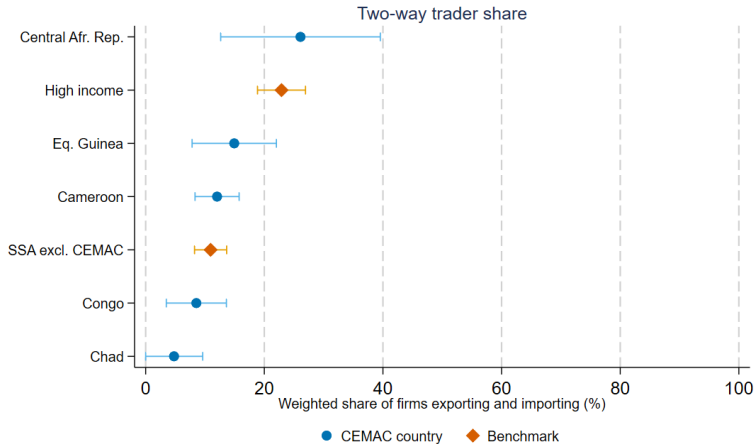
WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

Import and Foreign-Input Participation



WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

Two-Way Trader Status



WBES weights used. Benchmark points are equal-country averages. Gabon is coverage-only; rows with N<30 suppressed.

$$\ln(E_i) = \alpha + \beta \ln(X_i^+) + \gamma \text{Exporter}_i + \varepsilon_i \quad (4)$$

- Latest-wave WBES estimates are weighted cross-sectional associations.
- X_i^+ is export value, with log export value set to zero for non-exporters in the all-firm model.
- The exporter dummy separates the extensive margin from the positive-export-value slope.
- Rows with fewer than 30 observations are suppressed; benchmark rows are equal-country averages.

WBES Country Trade-Employment Estimates

	All-firm slope	SE	N	Exporters	Exporter-only slope	SE	N	Countries
Cameroon	0.363	0.034	615	138	0.363	0.034	138	1
Central Afr. Rep.	-0.128	0.090	134	46	-0.128	0.091	46	1
Chad	0.235	0.058	164	13	.	.	.	1
Congo	0.166	0.077	344	49	0.166	0.078	49	1
Eq. Guinea	0.433	0.105	172	18	.	.	.	1
CEMAC average	0.214	0.097	1,429	264	0.134	0.143	233	5
High income	0.341	0.018	30,199	10,818	0.333	0.015	10,753	45
SSA excl. CEMAC	0.281	0.034	17,271	2,808	0.302	0.025	2,627	38

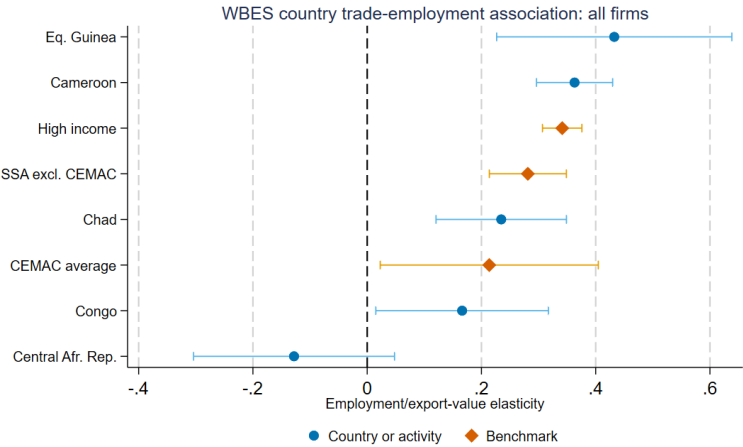
Slopes are weighted cross-sectional log employment/log export-value associations.

WBES Country Specification: All Firms

$$\ln(E_i) = \alpha_c + \beta_c \ln(X_i^+) + \gamma_c \text{Exporter}_i + \varepsilon_i \quad (5)$$

- Estimated separately by country using WBES weights.
- Non-exporters stay in the sample with $\ln(X_i^+) = 0$.
- The plotted coefficient is β_c ; it is descriptive, not causal.

WBES Country Associations: All Firms



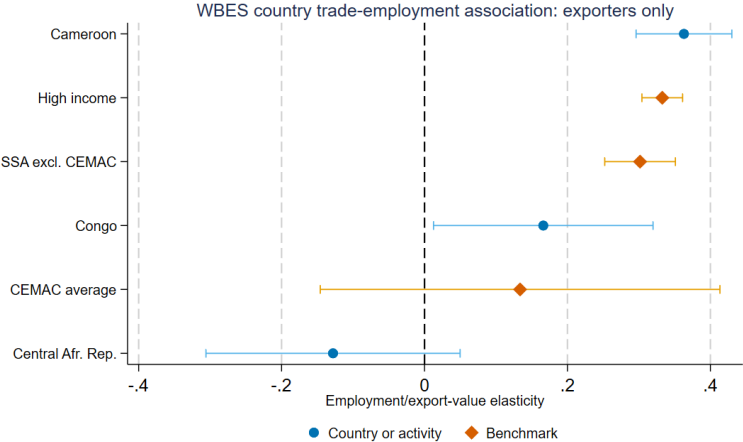
Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.

WBES Country Specification: Exporters Only

$$\ln(E_i) = \alpha_c + \beta_c \ln(X_i) + \varepsilon_i, \quad X_i > 0 \quad (6)$$

- Estimated separately by country among firms with positive export value.
- This is the intensive-margin association within exporters.
- Countries with fewer than 30 exporters are suppressed in the plot.

WBES Country Associations: Exporters Only



Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.

Cameroon WBES Activity Estimates

	All-firm slope	SE	N	Exporters	Exporter-only slope	SE	N
Construction/utilities	0.216	0.201	60	8	.	.	.
Manufacturing	0.546	0.064	161	53	0.546	0.065	53
Other services	0.373	0.037	52	9	.	.	.
Trade/hospitality/transport	0.300	0.049	342	68	0.300	0.050	68

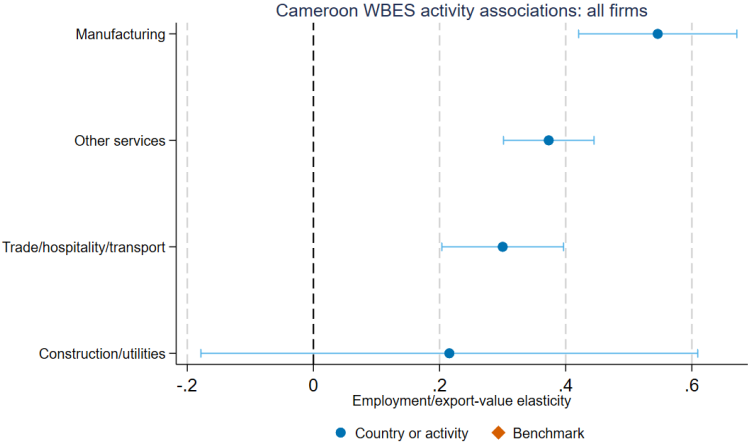
Activity groups use WBES-ISIC aggregates

Cameroon Activity Specification: All Firms

$$\ln(E_i) = \alpha_a + \beta_a \ln(X_i^+) + \gamma_a \text{Exporter}_i + \varepsilon_i \quad (7)$$

- Estimated within Cameroon WBES activity groups.
- This is a diagnostic adaptation of the elasticity idea to cross-sectional survey data.
- Activity rows pass the all-firm sample threshold, but exporter-only rows may be suppressed.

Cameroon Activity Associations: All Firms



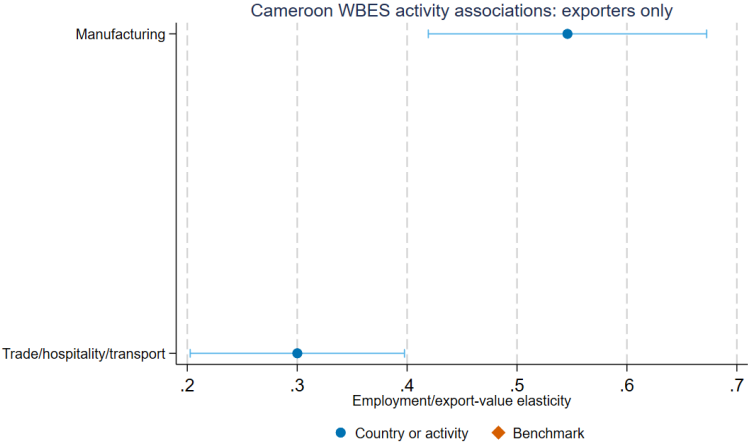
Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.

Cameroon Activity Specification: Exporters Only

$$\ln(E_i) = \alpha_a + \beta_a \ln(X_i) + \varepsilon_i, \quad X_i > 0 \quad (8)$$

- Estimated among Cameroon exporters within WBES activity groups.
- Rows are shown only when at least 30 exporter observations are available.
- These results are best read as sample diagnostics for follow-up, not sector multipliers.

Cameroon Activity Associations: Exporters Only



Weighted WBES cross-section. Points show log employment/log export-value associations; rows with N<30 suppressed.